

Strategy 1

Fair Price Mean Reversion

A session-anchored mean-reversion system trading BOS / CHoCH market-structure breaks back toward a session's Fair Price. Built for MNQ 1-minute, ported across three platforms.

INTERNAL NAME

FairPriceMeanReversion
(FPMR)

INSTRUMENTS

MNQ futures · NAS100 CFD

EXECUTION TIMEFRAME

1 minute

PLATFORMS

Pine v6 · NinjaTrader 8 · MetaTrader 5

CODEBASE

~9,900 lines across 41 source files

BACKTEST STATUS

PENDING RUN

1 Executive summary

Strategy 1 fades intraday extensions back toward a reference price fixed at the start of each trading session. It does not predict direction from candle colour, indicators or patterns — direction comes entirely from *which side of Fair Price the market is on*, and timing comes entirely from a market-structure break.

THE THREE-SENTENCE VERSION

At the open of each configured session the strategy records a **Fair Price** — the first 1-minute candle of that session. Once price has travelled far enough away from that price, it waits for a confirmed swing level to break (a **CHoCH** or **BOS**), and enters on that breaking candle in the direction that heads *back* toward Fair Price. The stop sits at the breaking candle’s extreme; the target is either a risk multiple or the Fair Price line itself, depending on how far out the entry was.

WHAT MAKES IT DISTINCTIVE

- **Direction is not a forecast.** Above Fair Price the strategy can only short; below it, only go long. There is no discretion.
- **Distance decides the exit.** Concentric percentage bands around Fair Price select between an R:R target and a target on the line itself.
- **Non-repainting by construction.** Swings are consumed only after their confirmation delay; Fair Price is read from a closed candle.
- **Every rejection is explained.** A 14-rung ordered ladder reports the single first gate that blocked each setup.

Where the project stands today

BUILD	STATUS	NOTES
Pine v6 reference spec	COMPLETE	The approved MVP and the behavioural contract the two ports are measured against. Superseded as the live build.
NinjaTrader 8 C# / NinjaScript	FEATURE COMPLETE	Compiles clean (csc.exe, exit 0). Carries the full news module. Backtest never executed — the Strategy Analyzer is GUI-only.
MetaTrader 5 MQL5 EA	PHASES 1–5	Most-tuned build; defaults were optimised on NAS100. News module (phase 6) is scaffolded but not implemented.

THE SINGLE MOST IMPORTANT OPEN ITEM

No backtest has ever been run on Strategy 1. Both `summary.json` and `trades.csv` in `backtest_results/2026/strategy_1/` are empty placeholders marked `PENDING_RUN`. Every performance claim about this strategy is currently unmeasured. The MT5 side has been observed running on 2025–2026 NAS100 data during bug hunts, but no results have been captured as a deliverable.

2 How the strategy functions

2.1 The per-bar pipeline

Everything runs as one pass per closed bar, in a fixed order. The order matters: each stage consumes the state the previous one produced, and the entry gates are evaluated in exactly this sequence so that the rejection reason is deterministic.

#	STAGE	WHAT IT DOES
1	Session management	Decides which of the three configured windows (if any) is active on this bar, in the chosen timezone.
2	Fair Price	On the second reference bar of a session, publishes Fair Price from the first bar. Fixed for the rest of that session.
3	Zone & bands	Computes the non-tradeable zone and the NEAR/FAR band edges as percentages of Fair Price.
4	Swing detection	Confirms pivot highs and lows using the left/right bar counts. A pivot becomes usable only after its right-bar delay.
5	Structure state	Maintains exactly one active high, one active low, and a bull/bear state flag.
6	Break detection	Tests the current bar against the active levels; classifies the break as CHoCH or BOS from the state at that instant.
7	Filters	Optional EMA and session-anchored VWAP gates.
8	Entry validation	The full gate ladder: session, Fair Price, warm-up, zone, side, caps, event type, filters, risk.
9	Order placement	Sizes the position and submits the entry with its bracket registered first.
10	Exit handling	Reads fills back from the platform, books P&L, applies trailing stops, updates daily accumulators.

2.2 Fair Price

Fair Price is *the* anchor of the whole system. For each enabled session it is the chosen price (Close on NT8, Open on MT5) of the **first 1-minute candle** of that session.

- It is published **one reference bar after** the session opens — reading the opening candle before it closes would be reading an unfinished bar, so the value appears at the close of bar 1 and becomes visible on bar 2.
- It **never changes** for the remainder of that session.
- Each enabled session gets its own; the previous session's value is discarded. (*MT5 adds an override: a session can be told to inherit an earlier session's Fair Price instead of setting its own.*)
- On NT8 a qualifying high-impact news release can replace it — see §6.

2.3 The zone and the distance bands

Concentric percentage bands around Fair Price decide both *whether* a setup may trade and *what it targets*. This replaced the original fixed ± 20 -point zone so that the geometry scales with price level.

TOO FAR	Beyond Band 2 — the move is over-extended, the setup is refused	rejected · TOO FAR
FAR BAND	Band 1 → Band 2 · SHORT on a qualifying break	TP = Fair Price
NEAR BAND	Zone edge → Band 1 · SHORT on a qualifying break	TP = risk × R:R
NO-TRADE ZONE	± zone % — no new entries; open trades are untouched	rejected · IN ZONE
FAIR PRICE		
NO-TRADE ZONE	± zone % — structure tracking continues normally inside it	rejected · IN ZONE
NEAR BAND	Zone edge → Band 1 · LONG on a qualifying break	TP = risk × R:R
FAR BAND	Band 1 → Band 2 · LONG on a qualifying break	TP = Fair Price
TOO FAR	Beyond Band 2 — the move is over-extended, the setup is refused	rejected · TOO FAR

The reasoning: close to Fair Price there is little room left to revert into, so a fixed risk multiple is the honest target. Further out there is a large gap to close, and aiming at the line itself captures it. Past Band 2 the move is treated as a trend rather than an extension, and nothing is taken. On MT5 the FAR band's Fair-Price target is switchable back to the R:R multiple.

2.4 Market structure — swings, active levels, CHoCH and BOS

The structure engine holds only **two floats**: one active high and one active low, each tagged with its role (HH/LH, HL/LL) and its confirmation bar index. There is no array of historical swings, so old structure physically cannot produce a signal.

STATE BEFORE BREAK	WHAT BREAKS	EVENT	NEW STATE
Bullish	Active low broken downward	Bearish CHoCH	Bearish
Bearish	Active low broken downward	Bearish BOS	Bearish
Bearish	Active high broken upward	Bullish CHoCH	Bullish
Bullish	Active high broken upward	Bullish BOS	Bullish

CHoCH is a break *against* the current structure state — a change of character. **BOS** is a break *with* it — a continuation. The candle that performs the break is the **displacement candle** by definition; nothing about its size, colour or body ratio is ever consulted. It is also the entry candle.

RULES THAT KEEP IT HONEST

- **A broken level is consumed.** Once broken it is cleared, so the same level can never fire twice and price sitting below a broken HL does not signal on every subsequent bar.
- **One side per bar.** A candle cannot break both sides; the bearish path is evaluated first and suppresses the bullish one.
- **Replacement, not accumulation.** When HL2 confirms, HL1 is gone — it is no longer stored anywhere. LatestSwing (default) accepts any newer confirmed pivot; RoleValidOnly requires the replacement to carry the role that matches the current state.
- **Setups expire.** An active level unbroken for SetupValidityBars bars since its *confirmation* bar is discarded.

- **A zone crossing resets everything.** If price flips to the other side of the Fair Price zone, the whole structure resets and seeds the state from the new side. A whipsaw across Fair Price wipes the active levels and the strategy waits for fresh pivots.

2.5 Entry models

A late addition that inverts the entire premise on a switch, so both readings can be tested from one description. Present on both NT8 and MT5.

MODEL	BEHAVIOUR
Reversion default	Above the zone shorts only , below it longs only . The trade heads back toward Fair Price. Both CHoCH and BOS are eligible, subject to their own switches. FAR-band setups target the Fair Price line.
BOS continuation	The inverse. Above the zone longs on a bullish BOS only ; below it shorts on a bearish BOS only — the move away from Fair Price is traded rather than faded. Two consequences are enforced rather than left to the switches: <i>every CHoCH is refused</i> regardless of “Take CHoCH entries”, because a change of character points back toward Fair Price; and a FAR-band setup takes the R:R target instead of Fair Price, which now sits behind a trade running away from it.

The active model is printed once at load, because it inverts the direction of every trade and that is not something to infer from the entry log.

3 Exits, sizing and risk control

3.1 Stop loss

The stop sits at the **displacement candle's extreme**, plus a configurable buffer: the candle's high for a short, its low for a long. It is submitted as an absolute price, not a distance from the fill, which is what makes the fill-slippage accounting in §7 meaningful. A `MinStopDistance` input rejects degenerate setups from very small displacement candles.

3.2 Take profit

MODE	TARGET
NEAR band	$\text{entry} \pm \text{risk} \times \text{RewardRatio}$ — default 1.5R.
FAR band	The Fair Price line itself. Sub-1R outcomes are accepted as-is; that is the intended trade-off of targeting the line. Skipped only if the target would land on the wrong side of the entry.
Fixed TP/SL master override	Ignores the structure stop, the band targets and the R:R entirely; every trade takes a fixed points stop and a fixed points target from the entry. On MT5 this is scoped to the near band and is on by default (30 pt stop, 48 pt target); on NT8 it is off.

3.3 Trailing stop

MODE	BEHAVIOUR
Off (default)	The stop stays where it was placed.
R-step	At +1R the stop moves to breakeven, at +2R to +1R, at +3R to +2R, and so on. R is the entry-to-initial-stop distance.
Structure	The stop trails confirmed swings — down to each lower swing high for shorts, up to each higher swing low for longs, reusing the same pivot and buffer settings as entries.

Both modes are ignored under Fixed TP/SL.

3.4 Position sizing

Sizing is **risk-based, not fixed-lot**. Quantity is chosen so the distance from entry to stop costs approximately the configured dollar risk, then bounded from both sides.

CONTROL	NT8 DEFAULT	MT5 DEFAULT	EFFECT
Risk target	\$100	500	What each trade aims to risk.
Risk tolerance	\$20	100	Reporting band only — does not change sizing.
Risk hard cap	\$150	1,000	Never exceeded. If one contract risks more than this, the trade is skipped with <code>RISK CAP</code> .
Max size	10 contracts	10 lots	Upper bound on quantity.

Worked example on MNQ with the NT8 defaults (point value \$2):

STOP DISTANCE	RISK / CONTRACT	QTY	RESULTING RISK	OUTCOME
10 pts	\$20	5	\$100	on target
20 pts	\$40	3	\$120	on target
60 pts	\$120	1	\$120	on target
75 pts	\$150	1	\$150	accepted — exactly at the cap
80 pts	\$160	—	—	skipped , logged RISK CAP
4 pts	\$8	10	\$80	clamped to max contracts

A CONSEQUENCE WORTH INTERNALISING

The hard cap skips *every* setup whose one-contract risk exceeds it — on MNQ, every displacement candle wider than 75 points. Wide displacement candles cluster in exactly the high-volatility conditions where the edge is least tested, so the backtest will never show what those trades would have done. The equity curve is a **filtered sample**; count the RISK CAP rejections to see how much was filtered out. Separately, losses are not uniformly sized: a 75-point stop risks 1.5× what a 10-point stop risks against the same target.

3.5 Daily P&L limits

A day-level circuit breaker layered on top of the per-trade sizing. It accumulates **realised, net** P&L — commissions included — and blocks further entries once a loss or profit limit would be reached.

- **Preventive, not reactive.** The gate does not merely block entries after the limit is breached; it refuses an entry when *realised P&L – open-trade risk – this trade’s risk* would breach it. Without this the day overshoots by the entire risk of the trade that broke the limit.
- Optionally flattens open positions when a limit is hit.
- **Off by default on NT8; on by default on MT5** (2,500 loss / 600 profit).

KNOWN LIMITATION, DELIBERATELY RETAINED

The limit is a bound on *intent*, not a guarantee. A stop can still fill through its level on a gap, so a day can finish beyond the limit even with the gate working perfectly.

3.6 Filters and the rejection ladder

Two optional filters, each provably inert when off (the condition is hard-coded true, so the trade list is byte-identical to a filters-off baseline):

- **EMA filter** — longs require $EMA1 > EMA2$, shorts require $EMA1 < EMA2$. Defaults 9 / 21.
- **VWAP filter** — session-anchored, accumulated in-strategy from $(H+L+C)/3 \times Volume$. Longs require close above, shorts below. Instruments with no volume yield NaN and the filter passes.

Every displacement candle that does not become a trade is tagged with the **first** gate that blocked it, evaluated in this exact order — so each has exactly one answer:

UNRECONCILED	SESSION	NO FP	WARMUP	IN ZONE	T00 FAR	SIDE	DAY CAP	SESS CAP	IN TRADE
EVT OFF	EMA	VWAP	RISK	RISK CAP					

4 Sessions, timezones and DST

Up to three independent windows, each with its own enable flag and HHMM–HHMM range (end exclusive). Overlapping windows resolve to the lowest-numbered enabled session. Each session sets its own Fair Price, resets structure, and resets its own per-session trade counter.

4.1 The IST anchoring scheme

One of the more consequential usability changes in the project. Windows are **typed in Asia/Kolkata as their summer (US-DST) values** and rebased once onto US Eastern, where they are fixed all year. In winter they therefore shift an hour later on the user's own clock automatically and keep tracking the same market hours.

The defaults were converted so the market hours traded are exactly unchanged: 0930–1000 ET → 1900–1930 IST. Verified by a unit test covering both seasons.

4.2 Two platform-specific hazards, both solved

- **NT8 — IANA ids do not exist on Windows.** `TimeZoneInfo.FindSystemTimeZoneById("America/New_York")` throws on the .NET Framework NinjaTrader runs on. A registry translates the Pine option list to Windows ids. An unresolvable zone is a *hard configuration error*: the strategy logs it and refuses to place orders rather than quietly falling back to machine local time.
- **NT8 — bar timestamps are close times**, where TradingView evaluates sessions against the bar's *open*. Session membership therefore uses `Time[0] - barLength`, so 0930–1000 means “bars that open at 09:30 through 09:59”, identical to Pine.
- **MT5 — the broker's server clock is arbitrary.** A dedicated input group declares the server's standard GMT offset and which DST calendar it follows, so session windows land correctly regardless of broker.

5 The news module — NinjaTrader 8 only

Deliberately deferred out of the Pine MVP: TradingView exposes no economic-calendar feed to scripts. It was built on NT8, where a ForexFactory export can be read from disk.

5.1 News Fair Price

When a qualifying high-impact release lands within a lookback window before a session opens, the **news candle** becomes that session's Fair Price reference instead of the normal first-candle rule. Configurable: file path, file timezone, CSV date format, lookback hours, impact filter, currency filter, and a rule for picking among multiple qualifying events (`First / Last / HighestImpact`).

THE JUDGEMENT CALLS, ALL DELIBERATE

- **The news candle becomes the session's logical start** when pre-session news trading is active. Structure resets there, the trade counter resets there, the warm-up starts there — and the clock open causes *no second reset*. Resetting twice would wipe the structure built during exactly the period the mode exists to trade.
- **Fair Price goes live on the news candle in both start modes.** Only *entries* are gated by the mode.
- **A missing candle is not the same as “no news”.** If a winning event is found but its candle is not in the loaded data, the strategy falls back to the first-candle rule and logs it, rather than trading a wrong reference price.
- **A missing or unparseable file disables the whole module** for the run, with an error in the log, and behaviour reverts to the Pine rule.

- With the toggle off, no calendar is read at all and no code path differs from Pine.

5.2 News surprise classification

A second layer that reads forecast / actual / previous values out of the calendar and classifies a release as *expected* or *unexpected* against configurable percentage thresholds, selecting the Fair Price reference accordingly. Ships with a **consolidation detector** (range-compression search) for locating the post-news balance area, and a rule for what to do when the numbers are missing. Off by default, and its parent toggle is off too, so default behaviour is unchanged.

HIGHEST-RISK INPUT IN THE WHOLE PORT

News-file timezone handling. JSON rows carry an explicit UTC offset and that offset wins. **CSV rows carry no zone at all** — their clock times are in whatever zone the exporting ForexFactory account was set to. Get it wrong and every news-session Fair Price is silently taken from the wrong candle. Verify against a known 08:30 ET release before trusting any news-enabled run.

6 Reverse mode — MetaTrader 5 only

A research instrument rather than a trading feature: it takes the opposite side of every signal the strategy generates, so the inverse of the edge can be measured directly. It grew from a boolean into a four-way mode as the difference between “opposite side” and “true P&L inverse” became clear.

MODE	BEHAVIOUR
Off (default)	Trade the setup as signalled.
Mirror bracket	Opposite side, same SL and TP <i>distances</i> . The reversed trade still has a near stop and a far target — so it is not the P&L inverse.
Swap bracket	Opposite side with SL and TP exchanged. The true inverse: the reversed trade loses exactly when the original would have won.
Swap, ORIGINAL size	As above, but the position keeps the size the original setup would have taken — the true dollar-for-dollar mirror. Ignores the risk cap and is warned about at load.

A REAL BUG THIS MODE EXPOSED

The preventive daily-loss gate was being handed the risk the position sizer *aimed for*, not the risk actually attached to the order. Under swap-with-original-size the position is sized on the signal’s stop but carries the far wider swapped stop, so the money genuinely at risk was **target/stop** times larger than the figure the gate checked. Measured over 2025 NAS100 with a 2,500 daily limit, that mode lost as much as **16,078 in a single New York trading day** and breached the limit on **35 of 231 days**. The gate now projects the risk of the stop actually on the trade. Every other mode has the two figures equal, so nothing else changed.

7 Where the platforms genuinely disagree

None of the items below are simplifications of the trading logic — the state machine, active-level rules, invalidation, sessions and filters are reproduced as specified. What differs is how each platform *executes*.

7.1 Entry fills at the next bar's open (NT8) — the largest divergence

Pine's `process_orders_on_close` fills the entry at the close of the signal candle, and the approved Pine results are built on that. NinjaTrader has no equivalent: with `Calculate.OnBarClose` the signal is generated after the bar closes and the market order fills at the **open of the next bar**. Using `OnEachTick` instead would allow an intrabar entry but would destroy the “closes through the level” definition of the displacement candle, which *is* the signal.

Because SL and TP are absolute prices computed from the displacement candle, a fill away from the signal price changes the realised R:R — so the strategy logs, on every fill, the slip in points, the actual dollar risk and the actual R:R.

READ THIS BEFORE COMPARING PLATFORMS

NinjaTrader's numbers will be worse than TradingView's, and the NinjaTrader numbers are the honest ones. A divergence here is not a bug. If you compare the two, the TradingView figures are the optimistic ones.

7.2 Other resolved divergences

AREA	RESOLUTION
Tick precision	Tick Replay is a data-series setting, not a script property, and is mutually exclusive with high fill resolution. <code>UseTickPrecision</code> instead sets <code>OrderFillResolution.High</code> at 1 tick — NinjaTrader's own “replay the real ticks through the fill engine”. This is the fix for same-bar TP/SL ambiguity that the TradingView build could not have.
Bracket orders	Managed <code>SetStopLoss/SetProfitTarget</code> with <code>PerEntryExecution</code> , registered <i>before</i> the entry. NinjaTrader handles OCO pairing, per-signal scoping, partial-fill resubmission and restart reconciliation. Exit reasons are read from the order that actually filled, not inferred from price.
Reference series	When the chart is already the reference timeframe, the primary series is reused and no extra data is loaded. The Fair Price engine is advanced by a cursor from inside the primary bar handler, not from <code>BarsInProgress == 1</code> — making results independent of multi-series call ordering, which silently differs between historical and real-time.
VWAP	NinjaTrader's bundled VWAP is order-flow licensed and anchors to the instrument's session. The spec wants an anchor at <i>your</i> session start, so it is accumulated in-strategy.
Restart / real-time	<code>StartBehavior.WaitUntilFlat</code> plus an explicit check on the historical→real-time transition. A live position with no matching trade record blocks entries and reports <code>UNRECONCILED</code> until flat; stale records against a flat account are dropped. Fills are never assumed.
Alerts	Pine's <code>alert()</code> became log output behind <code>VerboseLogging</code> ; hard errors go to the Log tab.
MT5 units	The input panel formerly mixed index points, ticks and MT5 points at once. On NAS100 a tick and an MT5 point are both 0.01 of an index point, so a minimum stop typed as 25 meant 0.25 and rejected nothing — a hundredfold silent difference from what the number looks like. Every distance is now an index point (which is exactly one MNQ point), and the EA states the unit with the live symbol's money value at startup.

8 Development progress

Eleven days of work, 30 commits, six merged pull requests — from the Pine MVP to two feature-matched platform builds.

30 Aug – 31 Aug 2026 · 1075fa3, 644aa0a

Phase 0 — Pine MVP as the baseline

The approved reference build: BOS/CHoCH/displacement structure engine, a fixed ± 20 -point Fair Price zone, three sessions, EMA and VWAP filters, 1.5R target, one-trade-at-a-time, and a full trade-visualisation engine. Critically, no first-candle-colour entry exists anywhere — the opening candle only ever sets Fair Price.

2 Sep 2026 · d19baaa · PR #1

News surprise classification, DST anchoring, backtest harness

The first substantial NT8 work. Added forecast/actual/previous parsing for both CSV and JSON, the surprise classifier, the consolidation detector, the `DstAnchor` mechanism, and the `backtest_results/` scaffold with its import scripts.

3 Sep 2026 · 1ca72ab · PR #2

Percentage zones, distance bands, trailing stops — the model reshaping

The most significant change to the trading logic since the MVP. The fixed-point zone became a percentage of Fair Price so the geometry scales with price level; the single zone became the NEAR/FAR/TOO-FAR band system that selects the exit target by distance; trailing stops (R-step and structure) were added; and pre-session news reversion was introduced. **This is where the extended-move TP override was retired** — the band system subsumes it.

4 Sep 2026 · 8b3d720 · PR #3

News Fair Price expiry toggle and a master news-trading switch

Control over whether a news-derived Fair Price expires at the session open, plus a single master toggle over all news behaviour.

4 Sep 2026 · 94cc1ae → dfc3134 · PR #4

MT5 port, phases 1–5

Sessions through filters ported to MQL5 as a 23-file EA. Then per-band exits, a 7–8:30 pm IST default session verified on NAS100, chart drawing of session boxes / Fair Price / trade zones, and the tuned settings adopted as the compiled defaults.

4 – 5 Sep 2026 · 1edc413 → 01bec5f · PR #5

MT5 hardening — reverse mode, Fair Price inheritance, unit unification

Reverse mode arrived as a boolean, became a three-way mode when “opposite side” proved not to be the P&L inverse, then a four-way with the original-size variant. The FAR band’s Fair-Price target was made switchable. Sessions gained the ability to inherit another session’s Fair Price. The daily-loss gate was fixed to project real rather than nominal risk. Finally, every distance input was unified to the index point, closing a silent 100× discrepancy.

5 – 6 Sep 2026 · ed60f4b, 0f8ab39 · PR #6

BOS-continuation entry model — both platforms realigned

The NinjaTrader half of the entry model MT5 already carried, so the two platforms stay configurable from one description. Reversion remains the default and is unchanged. Followed by further reverse-mode refinements. *Not yet backtested on either platform.*

8.1 The NinjaTrader change log in detail

Eight tracked changes, every one made *before* any backtest was run, each recording whether trading logic was affected — because that determines whether before-and-after results would be comparable. All verified against `csc.exe` (.NET Framework 4.8); NinjaTrader’s own compiler remains authoritative.

#	CHANGE	LOGIC IMPACT	DETAIL
1	Daily P&L limits added	YES — DEFAULT OFF	New group “5b”: loss limit, profit limit, flatten-on-limit, realised-P&L accumulation on exit fills, and two new entry gates.
2	Limit made preventive, not reactive	YES — WHEN ON	User reported the day’s net exceeded the limit. The gate only blocked entries <i>after</i> the breach, so the day overshot by the whole risk of the trade that broke it. Now projects headroom before entering.
3	Partial exit fills were over-booked	BUG FIX	The trade record was closed on the <i>first</i> exit execution and the full entry quantity booked at that partial’s price; remaining fills were ignored, and entry partials were truncated to the first fill. Now each execution books at its own price and quantity, entry fills accumulate to a volume-weighted average, and a record closes only when every filled contract is out.
4	P&L lost across a day boundary	BUG FIX	The accumulator reset in <code>OnBarUpdate</code> on a new day, but booking happens in <code>OnExecutionUpdate</code> , which fires first for that bar — so a trade exiting on the first bar of a new day was booked to the finished day and then wiped by the reset, lost from both. Now keyed to the trading day derived from the execution time.
5	Commission included in the daily net	LIMIT ONLY	Only gross price difference was booked. Entry and exit commissions are now booked as charged.
6	All chart visuals removed	NONE	Deleted <code>VisualEngine.cs</code> (383 lines) and stripped 16 call sites; per-bar <code>Draw.*</code> calls over ~30k bars/month were a measurable backtest cost. The state panel was replaced with a run summary printed at termination.
7	News surprise classification	YES — DEFAULT OFF	Forecast/actual/previous parsing, the classifier, the consolidation detector and group “6b”. Both this and its parent toggle are off by default.
8	Sessions moved to IST with auto DST	NO HOURS CHANGE	Windows typed as summer IST and rebased onto US Eastern. Defaults converted to preserve the exact same market hours; verified by unit test across both seasons.

9 Platform comparison

9.1 Feature parity

CAPABILITY	NINJATRADER 8	METATRADER 5	NOTE
Structure engine (CHoCH / BOS / displacement)	YES	YES	Identical rules
Percentage zone + NEAR/FAR bands	YES	YES	MT5 adds a FAR-band mode switch
Entry model (Reversion / BOS continuation)	YES	YES	Realigned in PR #6
Trailing stops (R-step / structure)	YES	YES	
Fixed TP/SL override	YES	YES	MT5 scopes it to the near band and defaults it on
Risk-based sizing + hard cap	YES	YES	Contracts vs lots
Daily P&L limits	YES	YES	MT5 defaults them on
EMA / VWAP filters	YES	YES	MT5 adds a VWAP volume-source mode
IST session anchoring + auto DST	YES	YES	MT5 adds broker server-clock handling
News Fair Price + surprise classifier	YES	PHASE 6	MT5 has the types and the engine's decision tree shaped for it, but no loader or resolver yet
Tick-precision fill resolution	YES	N/A	Platform-specific mechanism
Reverse mode (4-way)	NO	YES	Research instrument, MT5 only
Fair Price inheritance between sessions	NO	YES	MT5 session 3 inherits session 1 by default
On-chart drawing	REMOVED	YES	Deleted from NT8 as a backtest cost; MT5 retains a chart painter

9.2 Default divergence — the two builds are configured very differently

NinjaTrader still ships close to the conservative Pine defaults. MetaTrader ships the settings that were tuned on NAS100. This is worth knowing before comparing any results across the two.

SETTING	NT8 DEFAULT	MT5 DEFAULT
Sessions enabled	1 of 3	3 of 3
Session 1 window (IST)	1900–1930	1900–2100
Fair Price source	Close	Open
No-trade zone	0.10%	0.30%
Band 1 / Band 2	0.30% / 0.60%	0.50% / 0.80%
Pivot left / right	3 / 2	1 / 1
Max trades per day	3	30
Max trades per session	0 (unlimited)	10
Only one open trade	true	false
Fixed TP/SL	off	on – 30 pt SL / 48 pt TP
Risk target / hard cap	\$100 / \$150	500 / 1,000
Daily P&L limits	off	on – 2,500 loss / 600 profit

READING THE DIVERGENCE

The MT5 defaults describe a materially more aggressive system: far looser pivots (1/1 confirms a swing almost immediately), ten times the daily trade allowance, concurrent positions permitted, five times the per-trade risk, and a fixed 30/48-point bracket in place of the structure stop. These are tuned values, not a second opinion about the strategy — but the two builds are not currently running the same experiment.

10 Backtesting — status and what to verify

10.1 Current status

FIELD	VALUE
Status	PENDING_RUN
Reason	The NinjaTrader 8 Strategy Analyzer is a GUI application with no headless or CLI mode, so it could not be driven programmatically.
Planned instrument	MNQ, MergeNonBackAdjusted, CME US Index Futures ETH
Planned range	2026-01-01 → 2026-09-02 (200 distinct days with 1-minute data verified present)
Timeframe	1 Minute, fill resolution High / Tick / 1
Contracts on disk	MNQ 03–26, MNQ 06–26, MNQ 09–26
Compile state	Clean — no compiler errors, exit 0
Next step	Run per backtest_results/README.md, export, then scripts/import_nt8_results.py populates summary.json.

10.2 Verify these three things on the first run

1. **The sizing arithmetic against the real point value.** Run verbose and read the first few entry lines. On MNQ, `PointValue` must print as **2.0**. If it prints anything else the instrument definition is wrong, not the code — and every figure in the sizing table is void.
2. **News-file timezone.** Only if news is enabled. Find a day with a known 08:30 ET release and confirm the logged Fair Price timestamp matches. An offset by a whole number of hours means the file zone is wrong. Also check the skipped-row count: a large one means the date format was mis-detected.
3. **Session windows against the bar timezone.** Confirm the session activates on the bar you expect. If the sessions are offset by hours, set the bar-timezone override. Checked once, then never again.

10.3 Where the backtest will read optimistically

Listed most to least material. These are known and accepted, not defects.

EFFECT	WHY IT FLATTERS THE RESULT
Sizing survivorship	The risk cap skips every setup wider than ~75 points on MNQ. Those cluster in high-volatility conditions where the edge is least tested, so the curve is a filtered sample.
Non-uniform loss sizes	A 60-point stop risks \$120 and a 75-point stop \$150 against a \$100 target. A run of wide-stop losers draws down faster than “three trades a day at \$100” suggests.
Slippage is zero by default	Set at least one tick for MNQ before drawing any conclusion. The entry is a market order at the next bar’s open — exactly where slippage lands.
Stop fills assume the stop price	Live, a gap through the stop fills worse. These stops sit at the displacement candle’s extreme — precisely where resting stops cluster.
The bar-close entry model	Costs one bar of gap risk on every trade (§7.1).
Same-bar TP/SL without tick precision	Turn tick precision off and same-bar outcomes revert to an assumption. If the same-bar-hit count is a meaningful fraction of trades, only the tick-precision run is worth reading.
News treated as known in advance	Release <i>times</i> genuinely are published ahead, so this is not look-ahead on price — but a file exported after the fact reflects the final schedule including mid-week reschedules. Use a file exported before the test period.
Limit fills require trade-through	<code>IsFillLimitOnTouch = false</code> — the conservative setting, left on deliberately. This one works <i>against</i> the result.

11 Current state, open items and file map

11.1 Open items

	ITEM	DETAIL
1	No backtest has been run	The blocking item. Nothing about this strategy’s performance is currently measured on either platform.
2	MT5 news module unimplemented	Phase 6. The shared types exist and the Fair Price engine’s decision tree is already shaped around them, so it is a resolver to fill in behind an interface that already fits — not a rewrite.
3	docs/PARAMETERS.md is stale	It still documents the fixed 20-point zone and the extended-move TP override, both retired in the 3 Sep band-system change, and predates fixed TP/SL, trailing stops, daily limits and the entry model. It no longer describes the shipping defaults.
4	Two files exceed the 500-line rule	<code>FairPriceMeanReversion.Trading.cs</code> is 704 lines against the project’s own 500-line limit; <code>Params.cs</code> is 479 and close behind.
5	NT8 and MT5 defaults have diverged	See §9.2. Fine for independent tuning, but the two builds are not currently comparable head-to-head.
6	NinjaScript compile not yet authoritative	Verification so far is via <code>csc.exe</code> against the NinjaTrader assemblies. NinjaTrader’s own F5 compile remains the authority and has not been run.

11.2 File map

NINJATRADER 8 — 22 FILES, ~4,875 LINES

FairPriceMeanReversion.cs	370
.Params.cs	479
.Setup.cs	392
.Trading.cs	704
FPMR/Core/ 13 files	1,529
FPMR/News/ 5 files	1,220

Core holds the session manager, Fair Price engine, structure engine, pivot detector, setup bands, risk sizer, VWAP, DST anchor, timezone registry, rejection reporter, trade record and consolidation detector. News holds the calendar loader, a minimal JSON parser, the event model, the Fair Price resolver and the surprise classifier.

METATRADER 5 — 23 FILES, ~5,222 LINES

FairPriceMeanReversion.mqh	349
FpmrStrategy.mqh	499
FpmrStrategyImpl.mqh	488
TradeManagerImpl.mqh	463
FpmrTimeZone.mqh	430
TradeManager.mqh	385
FpmrEnums.mqh	321
ChartPainter.mqh	267
15 further includes	2,020

The include set mirrors the NT8 Core folder one-for-one, plus a chart painter and a symbol-abstraction layer that normalises index points across NAS100 and MNQ.

SUPPORTING MATERIAL

Product_Requirements/*.pine, *_DOCS.md	The Pine MVP and its full specification — the behavioural contract.
docs/DESIGN_DECISIONS.md	Every place NinjaTrader forced a deviation, ranked by how much it moves results.
docs/BACKTEST_NOTES.md	What to verify first, and where the backtest reads optimistically.
docs/PARAMETERS.md	Full parameter reference — currently stale , see §11.1.
backtest_results/2026/strategy_1/	Config, change log, error log and empty result placeholders.
scripts/deploy_to_ninjatrader.sh	Mirrors the source tree into the NinjaTrader Custom\\Strategies folder.

IN ONE LINE

Strategy 1 is a complete, carefully-documented, twice-ported trading system with a well-reasoned set of gates and an unusually honest account of its own limitations — and it has never been measured. Running the 2026 MNQ backtest is the one action that would change what is known about it.